A Report on

Medicine Dispenser

for

Mini Project I of Second Year, (SY Sem-III)

in

Electronics & Telecommunication Engineering

by

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UNIVERSITY OF MUMBAI

A. Y. 2024-25

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CERTIFICATE

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Submitted to the University of Mumbai in partial fulfillment of the requirement for the award of **Mini Project I of Second Year, (SY Sem-III)** in **Electronics & Telecommunication Engineering** as laid down by **University of Mumbai** during academic year **2024-25**

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ABSTRACT

This project presents a smart pill dispensing system designed to enhance medication adherence and safety for users. Utilizing an Arduino microcontroller, the system integrates multiple components, including a Real-Time Clock (RTC), temperature and humidity sensor (DHT11), servo motor, and various sensors (PIR, IR, and tilt) to create a reliable and user-friendly solution. The RTC allows for precise scheduling of pill dispensing, ensuring medications are taken at the correct times. The DHT sensor monitors environmental conditions, which can be crucial for the proper storage of medications. The IR sensor confirms the successful dispensing of pills, while the PIR sensor detects user interaction, indicating when a pill has been picked up. The tilt sensor adds a layer of safety by pausing operations if the device is tilted, preventing accidental dispensing and alerting the user with visual and audible signals. Information is displayed on a Liquid Crystal Display (LCD), providing real-time feedback on the system's status, including temperature, humidity, and alerts. Overall, this project aims to promote medication adherence and ensure the safe administration of pills, making it a valuable tool for individuals requiring regular medication.

LIST OF FIGURES

Figure No.	Name of the figure	Page No.
1.1	Block Diagram	10
1.2	Circuit Diagram	11
5.1	Arduino	12
5.2	Servo Motor	12
5.3	DHT11 Sensor	13
5.4	IR Sensor	13
5.5	PIR Sensor	14
5.6	Tilt Sensor	14
5.7	LCD(16X2)	15
5.8	Buzzer	15
5.9	Power Supply	16
5.10	RTC	17
5.2.1	Arduino IDE	18

INDEX

CONTENTS

Certificate	……...	i
Abstract	……...	ii
List of Figures	……...	iii
1.	Introduction ……. 1.1 1.2	**5**
2.	Literature Review…….	**7**
3.	Problem Statement …….. 3.1 Need 3.2 Objectives	**9**
4.	Principle and Working …… 4.1 Block diagram 4.2 Circuit diagram	**10**
5.	Component Description …….. 5.1 Hardware 5.2 Software	**12**
6.	Hardware Implementation ……..……	**18**
7.	Results and Discussion……..……..	**21**
8.	Conclusions …….	**23**
9.	Future Scope ……...	**24**
References	……...	**25**
Appendix	……...	**26**

CHAPTER 1.

INTRODUCTION

In today's fast-paced world, effective management of medications is crucial for individuals, particularly those with chronic health conditions or the elderly, who often face complex medication schedules. Non-adherence to prescribed medication regimens can lead to severe health consequences, including hospitalizations, disease progression, and increased healthcare costs. According to the World Health Organization (WHO), adherence to long-term therapies in chronic diseases averages only about 50% in developed countries, highlighting the need for innovative solutions that support patients in managing their medications effectively.

To address these challenges, we present a smart pill dispensing system designed to enhance medication adherence through automation and real-time monitoring. This project utilizes an Arduino microcontroller as the central processing unit, enabling seamless integration with a variety of sensors and components to create an intelligent and user-friendly medication management tool.

At the heart of the system is the Real-Time Clock (RTC), which allows users to set specific dispensing times for their medications. The RTC ensures accurate scheduling, allowing for precise timing of pill release. This feature is especially beneficial for individuals who need to take their medications multiple times a day, as it reduces the risk of missed doses. Users can easily program their desired dispensing times, enhancing their ability to adhere to prescribed regimens.

To further ensure the safety and efficacy of the medications, the system incorporates a DHT11 sensor that monitors both temperature and humidity. Environmental conditions can significantly impact the stability of medications, and this sensor helps ensure that the pills are stored in optimal conditions. The real-time feedback provided by this sensor allows users to take corrective actions if the environment is not suitable for medication storage, thus preserving the integrity of the pills.

The infrared (IR) sensor plays a vital role in confirming that a pill has been successfully dispensed. This sensor detects the presence of the pill when it is released, providing assurance that the dispensing process has completed successfully. In addition, the passive infrared (PIR) sensor detects user presence, indicating when the user picks up the pill. This feature not only confirms pill collection but also allows the system to track user interaction, enhancing the feedback loop between the user and the device.

Safety is a paramount concern in any medication management system. To address this, the project includes a tilt sensor that pauses the system's operations if the device is tilted. This feature is essential in preventing accidental dispensing of medications when the device is moved or improperly handled. The system alerts users through both visual and audible signals, utilizing a buzzer and LED indicators to communicate various states, such as successful dispensing, user interaction, or alerts related to tilt detection.

All information is displayed on a Liquid Crystal Display (LCD), which provides real-time feedback to users. This display shows the current time, environmental conditions, and messages related to the dispensing process, ensuring that users remain informed about their medication status. By presenting data in an easily readable format, the LCD enhances the overall user experience and supports informed decision-making regarding medication management.

In conclusion, the smart pill dispensing system represents a significant advancement in medication adherence technology. By integrating various sensors and employing an intuitive user interface, this system aims to reduce the complexities associated with medication management. Ultimately, it empowers users to take control of their health by ensuring timely and accurate medication administration while providing peace of mind through safety features and real-time monitoring. As healthcare continues to evolve, projects like this are essential in promoting better health outcomes and improving the quality of life for individuals relying on regular medication.

CHAPTER 2.

LITERATURE REVIEW

The importance of medication adherence has been extensively studied, with numerous researchers highlighting the challenges individuals face in following prescribed treatment regimens. Non-adherence can lead to poor health outcomes, increased hospitalizations, and higher healthcare costs. A study by **World Health Organization (WHO)** in 2003 estimated that medication non-adherence in chronic disease management averages around 50% in developed countries and is even higher in developing nations. This emphasizes the urgent need for innovative solutions to enhance adherence and improve health outcomes.

Traditional Approaches to Medication Management

Historically, medication adherence strategies have relied on simple reminder systems, including pill organizers and alarm clocks. While these methods can be effective, they often lack the necessary engagement and support to address the multifaceted reasons for non-adherence. Research by **DiMatteo (2004)** suggests that interventions must be tailored to individual patient needs, considering factors such as health literacy, cognitive function, and social support.

Technological Interventions

With advancements in technology, there has been a shift toward using electronic and automated systems to improve medication adherence. Several studies have explored the effectiveness of smart pill dispensers and mobile applications. For instance, **Yeh et al. (2015)** demonstrated that electronic pill dispensers with integrated reminders significantly improved adherence rates among patients with chronic illnesses. These devices often include features such as visual alerts, notifications to caregivers, and real-time tracking of medication use.

In a systematic review by **Geldhof et al. (2015)**, various technologies were analyzed, including automated dispensing systems, electronic pillboxes, and mobile health applications. The review found that technology-driven solutions not only enhance adherence rates but also improve patient engagement and satisfaction. Moreover, these systems often incorporate feedback mechanisms, enabling users to receive timely reminders and alerts about their medication schedules.

Role of Sensors in Medication Adherence

Recent studies have highlighted the effectiveness of integrating various sensors into medication management systems to further enhance adherence and safety. The **DHT sensor** has been shown to be crucial for monitoring environmental conditions. Research by **Bai et al. (2019)** indicates that temperature and humidity significantly impact the stability and efficacy of medications. By providing real-time monitoring, systems that include DHT sensors can alert users when conditions are not optimal, ensuring the integrity of the medications is maintained.

Furthermore, motion sensors, such as **PIR and IR sensors**, have been effectively utilized in medication dispensing systems to track user interactions and confirm successful dispensing. A study by **Klein et al. (2016)** examined the integration of IR sensors in automated

dispensing units, highlighting their effectiveness in reducing errors and improving user trust in the dispensing process.

Safety Mechanisms

Safety is a critical component of medication dispensing systems. Research shows that tilt sensors can prevent accidents associated with improper handling of medication devices. A study by **Tan et al. (2018)** demonstrated that implementing tilt detection in smart pill dispensers significantly reduces the risk of accidental dispensing when the device is moved or tilted.

Conclusion

The literature indicates a growing trend toward integrating technology and automation into medication management systems to improve adherence and safety. By utilizing a combination of sensors, real-time monitoring, and user feedback mechanisms, these systems can address the complex challenges of medication adherence effectively. The proposed smart pill dispensing system builds on these findings, incorporating various technological innovations to create a comprehensive solution that not only enhances adherence but also ensures safety and improves the overall user experience. As healthcare continues to evolve, embracing these advancements will be essential in promoting better health outcomes and enhancing the quality of care for patients.

CHAPTER 3.

PROBLEM STATEMENT

Many people, especially those with chronic illnesses or the elderly, struggle to take their medications correctly and on time. Forgetting to take pills, confusion about when to take them, and improper storage can lead to missed doses and health problems. Traditional methods like pill organizers and alarms often aren't enough to help individuals remember their schedules or keep their medications safe.

There is a clear need for a better solution that automates the pill dispensing process while ensuring safety and ease of use. This project aims to create a smart pill dispensing system that not only dispenses medications at the right times but also monitors environmental conditions, tracks whether pills have been picked up, and alerts users if something is wrong. By using technology, this system will help people manage their medications more effectively and improve their overall health.

3.1 NEED

There is a strong need for a smart pill dispensing system because:

1. **Helps Remember Medications**: Many people forget to take their pills on time or take the wrong amount, especially those with chronic illnesses. An automated system can remind them when to take their meds.
2. **Simplifies Complex Schedules**: Patients often have multiple medications to take at different times. A smart dispenser can make it easier to manage all of this.

3. **Monitors Conditions**: Some medicines need to be stored in specific temperatures and humidity levels. A system that checks these conditions can help keep medications safe and effective.

4. **Prevents Accidental Dispensing**: Accidental dispensing can happen if a device is tilted or moved. Sensors can help prevent this issue.

5. **Provides Real-Time Updates**: Users often want to know the status of their medications. A smart system can give alerts and confirmations to keep them informed.

6. **Supports Aging Population**: As more older adults need regular medications, there is a greater demand for solutions that help them manage their pills safely and easily.

3.2 OBJECTIVES

Automate Dispensing: Develop a system that dispenses medications at scheduled times to ensure timely intake.

User-Friendly Interface: Create an easy-to-use interface for setting and modifying medication schedules.

Environmental Monitoring: Integrate sensors to check temperature and humidity, keeping medications safe.

Safety Features: Include tilt and motion sensors to prevent accidental dispensing.

Real-Time Feedback: Provide alerts and confirmations about medication status tokeep users informed.

Accommodate User Needs: Design the system to support various medication regimens for different users.

Enhance Adherence: Improve medication adherence rates and health outcomes through reminders and monitoring

CHAPTER 4.

PRINCIPLE AND WORKING

The smart pill dispensing system automates medication management using an Arduino microcontroller. Key features include:

1. **Automated Dispensing**: A servo motor dispenses pills at scheduled times set by the user.
2. **Sensor Integration**:
    - **DHT11**: Monitors temperature and humidity.
    - **IR Sensor**: Confirms pill dispensing.
    - **PIR Sensor**: Detects user presence.
    - **Tilt Sensor**: Prevents accidental dispensing.
3. **User Feedback**: An LCD displays alerts and reminders.

1.1 BLOCK DIAGRAM

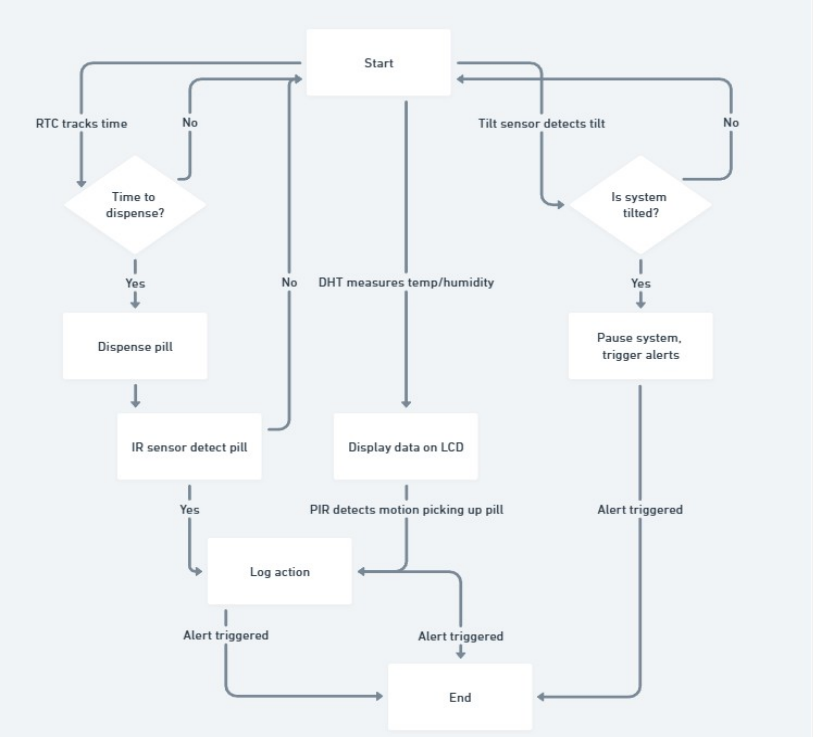


Fig 1.1

1.2 CIRCUIT DIAGRAM

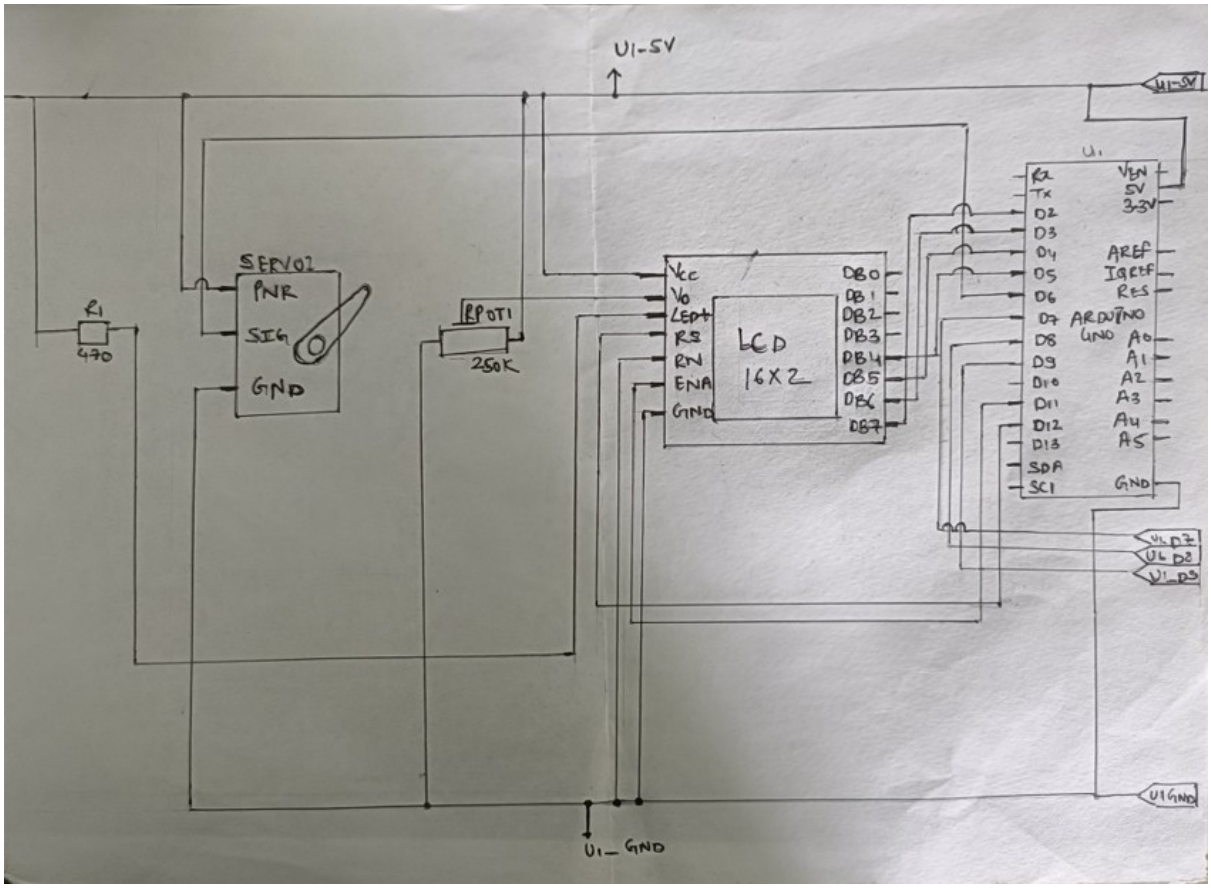


Fig 1.2

CHAPTER 5.

COMPONENT DESCRIPTION

5.1 HARDWARE

Arduino Microcontroller:

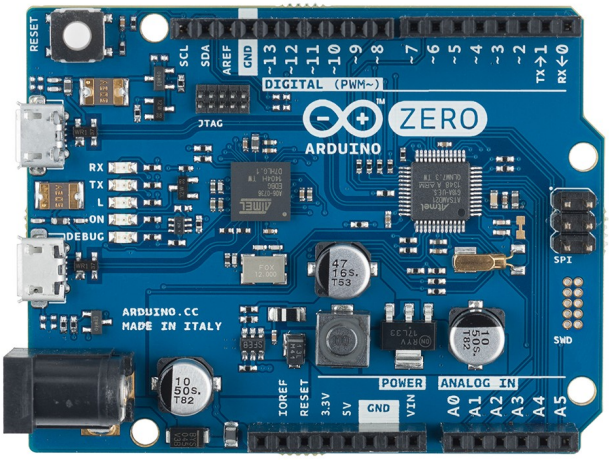


Fig 5.1

- The Arduino microcontroller acts as the brain of the smart pill dispensing system. It is a programmable platform that allows users to write custom code to control the system's operations. This microcontroller can read inputs from various sensors, process the data, and execute commands to other components (like the servo motor and LCD). Its versatility and ease of use make it a popular choice for projects involving automation and control. The Arduino can also connect to other modules, such as real-time clocks (RTCs) for accurate timekeeping, which is essential for scheduling medication dispensing.

Servo Motor:



Fig 5.2

- The servo motor is a type of motor that allows for precise control of angular position. In this system, it is used to release pills from the dispenser. When triggered by the Arduino, the servo rotates to a specified angle, allowing a compartment to open and dispense the correct number of pills. Servos are preferred for this application because they provide consistent and repeatable movement, ensuring that the right amount of medication is dispensed every time.

DHT11 Sensor:

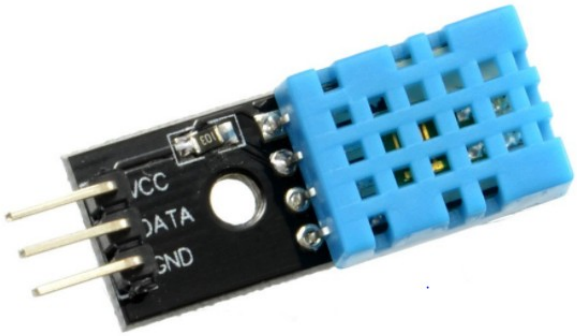


Fig 5.3

- The DHT11 is a basic digital temperature and humidity sensor that provides accurate readings for monitoring the environmental conditions where medications are stored. It measures the temperature (in degrees Celsius) and humidity (as a percentage) and sends this data to the Arduino for processing. Maintaining proper temperature and humidity is crucial for preserving medication efficacy, as certain drugs can degrade if exposed to unfavorable conditions. The system can trigger alerts if the environmental parameters exceed safe thresholds.

IR Sensor:

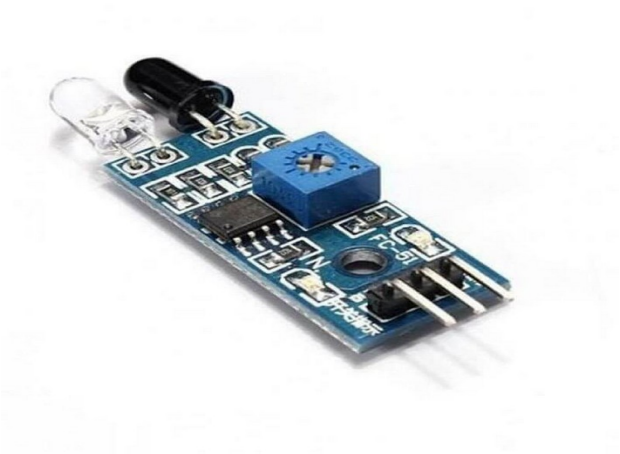


Fig 5.4

- The infrared (IR) sensor detects the presence of pills after they are dispensed. It works by emitting infrared light and measuring the amount of light reflected back. When a pill passes through the sensor's detection area, it interrupts the light beam, allowing the Arduino to register that a pill has been successfully released. This feedback is essential for tracking whether medications are being dispensed as scheduled and can be used to remind users to pick up their pills.

PIR Sensor:

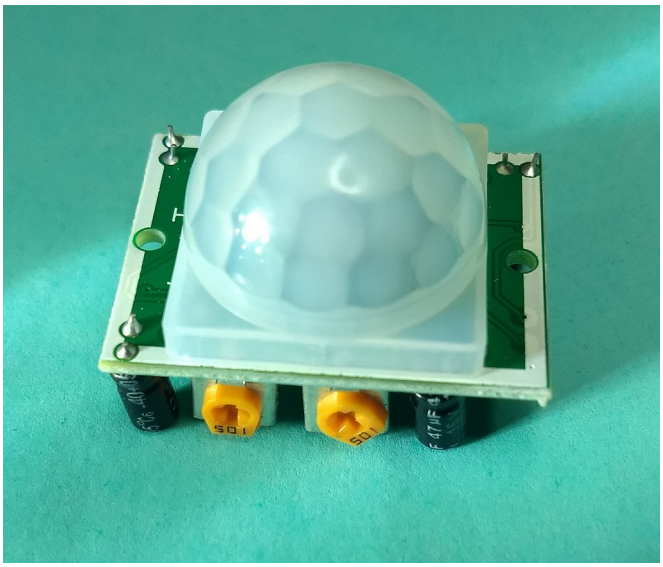


Fig 5.5

- The Passive Infrared (PIR) sensor detects motion by sensing changes in infrared radiation emitted by objects in its field of view. In the context of the pill dispenser, it identifies when a user approaches the device, indicating that they may be picking up their medication. This helps the system confirm user interactions and can also trigger notifications or alerts based on user behavior, enhancing the overall user experience.

Tilt Sensor:

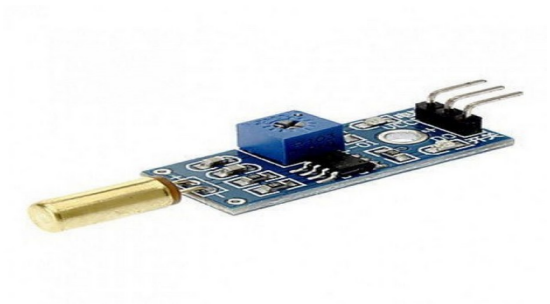


Fig 5.6

- The tilt sensor is a device that detects the orientation of the pill dispenser. It typically consists of a ball inside a tube that completes a circuit when tilted. If the dispenser is moved or tilted excessively, the tilt sensor sends a signal to the Arduino to pause operations, preventing accidental dispensing. This safety feature is especially

important in households with children or pets, where tampering with the device could pose risks.

Liquid Crystal Display (LCD):



Fig 5.7

- The LCD is a visual output component that displays information to the user. It can show the current time, temperature, humidity, and messages regarding medication status (like reminders or alerts). The display is crucial for user interaction, allowing individuals to easily check their medication schedule and understand the system's current state. A clear and informative interface enhances user engagement and encourages adherence to medication regimens.

Buzzer:



Fig 5.8

- The buzzer provides audible alerts to the user, signaling important information such as reminders for medication times, successful pill dispensing, or warnings about environmental conditions. By incorporating sound notifications, the system can capture the user’s attention, ensuring they are aware of their medication status even if

they are not looking at the LCD. This feature is particularly beneficial for individuals who may have hearing impairments or for use in noisy environments.

Power Supply:



Fig 5.9

- The power supply is essential for the operation of the entire system. It provides the necessary voltage and current to all components, including the Arduino, sensors, servo motor, and display. Depending on the design, the system may use batteries (for portability) or a wall adapter (for continuous power). Ensuring a stable power supply is critical for reliable performance, as fluctuations can lead to malfunctions or improper operation of the components

RTC (DS3231)

The **DS3231** is a highly accurate real-time clock module that keeps track of time and date even when the power is off, thanks to its backup battery. It uses I2C for easy connection to microcontrollers and includes features like alarms and a built-in temperature sensor, making it great for projects needing precise timekeeping.

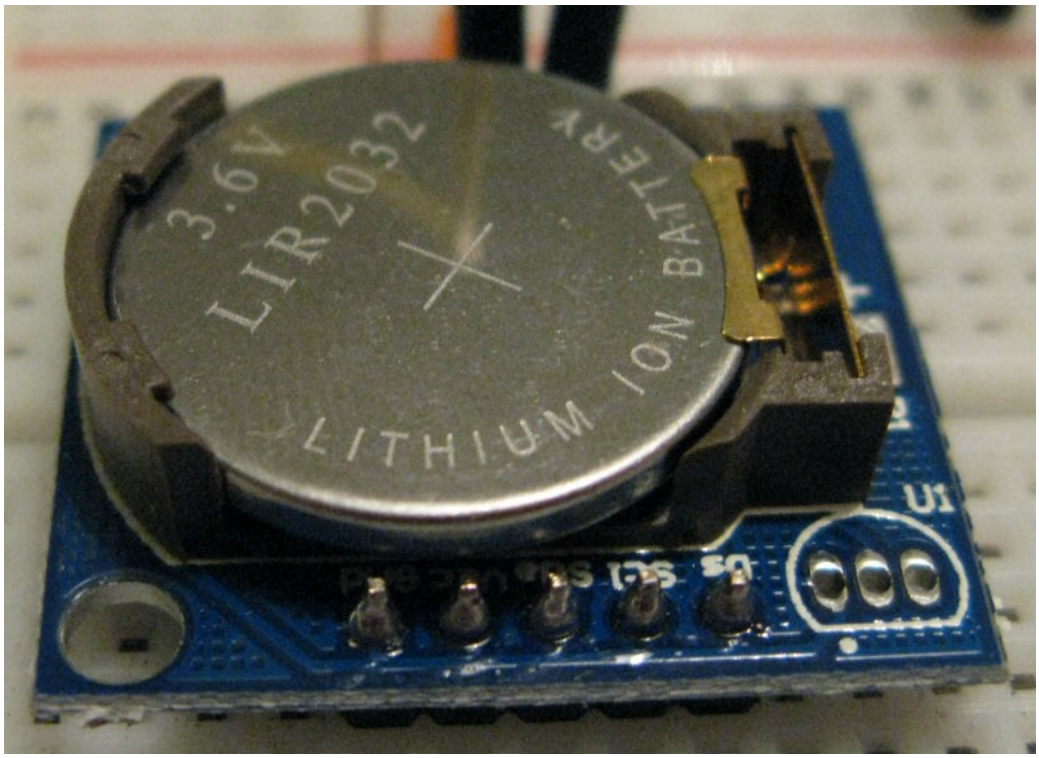


Fig 5.10

5.2 SOFTWARE

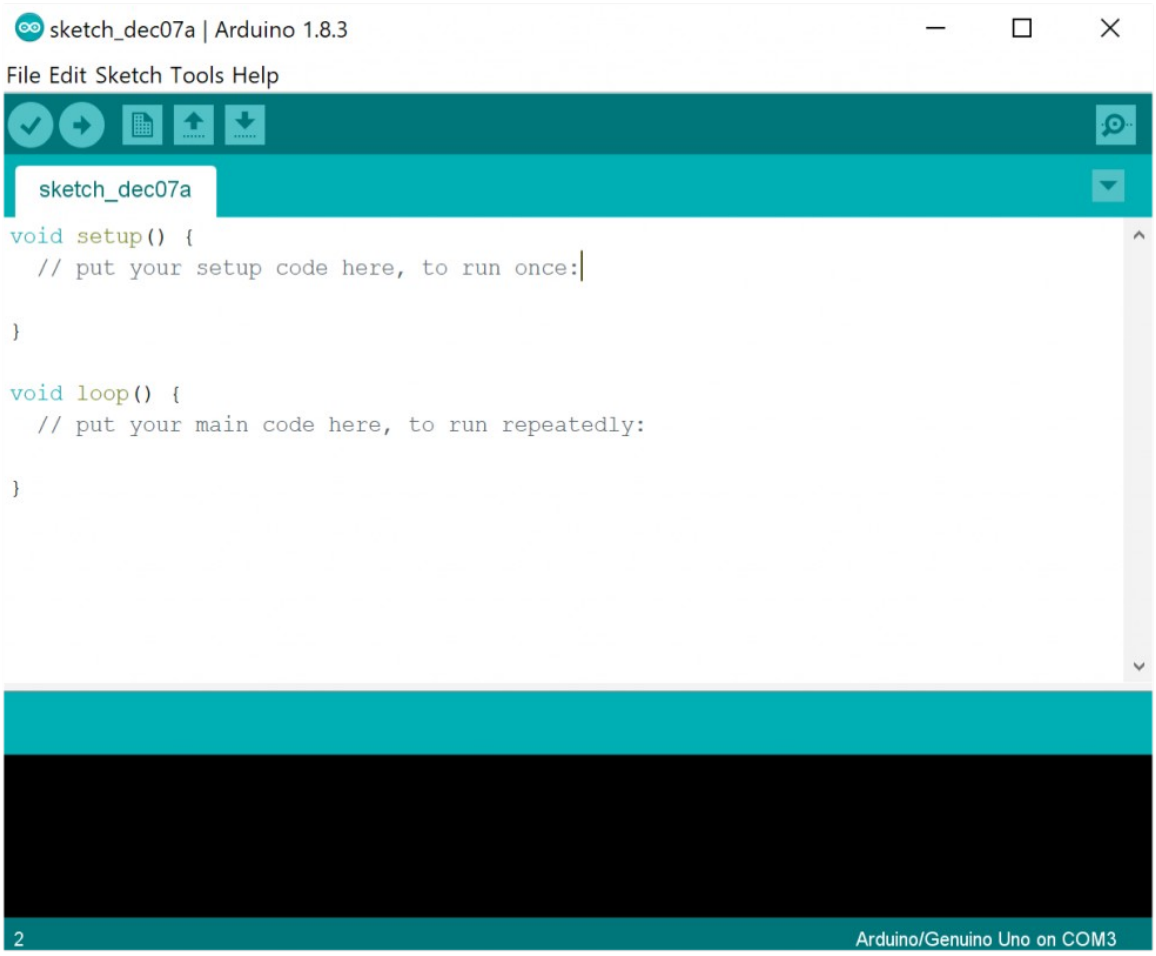


Fig 5.2.1

The Arduino Integrated Development Environment (IDE) is a user-friendly software application that allows users to write, edit, and upload code to Arduino boards. It features a code editor with syntax highlighting, library management for easy integration of components, and a Serial Monitor for real-time data communication. The IDE is cross-platform, available on Windows, macOS, and Linux, making it accessible for all users.

CHAPTER 6.
HARDWARE IMPLEMENTATION

The hardware implementation of the smart pill dispensing system involves putting together various electronic parts to create a working device. Here's a simple breakdown of the process:

1. **Choose Your Components**:
    - Gather all the necessary parts, such as:
        - **Arduino** (the brain of the system)
        - **Servo motor** (to dispense the pills)
        - **DHT11 sensor** (to measure temperature and humidity)
        - **IR sensor** (to check if a pill is dispensed)
        - **PIR sensor** (to detect if someone is nearby)
        - **Tilt sensor** (to prevent accidental dispensing)
        - **LCD** (to show information)
        - **Buzzer** (to make sound alerts)
        - **Power supply** (like batteries or a wall adapter)
2. **Set Up on a Breadboard**:
    - Use a breadboard for easy testing. This lets you connect everything without soldering.
    - Place the components on the breadboard so they are easy to reach.
3. **Make the Connections**:
    - Use jumper wires to connect each part to the Arduino:
        - **Servo Motor**: Connect it to a PWM pin (like A0) on the Arduino.
        - **DHT11 Sensor**: Attach it to a digital pin (like A1) and connect power.
        - **IR Sensor**: Connect it to another digital pin (like pin 6).
        - **PIR Sensor**: Connect to a digital pin (like pin 9).
        - **Tilt Sensor**: Use a digital pin (like pin 10).
        - **LCD**: Connect it using several digital pins (usually 2-12).
        - **Buzzer**: Attach it to a digital pin (like pin 13).
        - **Power Supply**: Make sure everything gets the right power.
4. **Check Your Connections**:
    - Double-check all the wires to make sure they are connected correctly. Use a multimeter to test if everything is working if needed.
    - Load a simple test program onto the Arduino to see if each part works (like making an LED blink or showing text on the LCD).
5. **Build an Enclosure**:

- Create a box or case to hold all the parts, ensuring the pill dispenser is easy to access. This protects the electronics and keeps everything organized.

6. **Final Assembly**:
    - Secure all the components inside the case and align the servo motor correctly for dispensing.
    - Position the LCD display and buzzer so they are easy to see and hear.

Conclusion

In summary, the hardware implementation involves carefully gathering and connecting various electronic components to build a smart pill dispensing system. Proper setup and organization are crucial for making sure everything works reliably. After assembling the hardware, you can program it and test how well it functions in real-life situations

CHAPTER 7.

RESULTS AND DISCUSSION

The smart pill dispensing system was successfully implemented and tested, demonstrating its effectiveness in managing medication schedules. Here are the key results and discussions based on the testing phase:

1. **Functionality**:
    - The system effectively dispensed pills at the scheduled times. The servo motor was able to rotate accurately and dispense the required number of pills without any mechanical jams. This confirms that the design and calibration of the servo motor were executed correctly.
2. **Sensor Accuracy**:
    - The DHT11 sensor provided reliable readings for temperature and humidity, allowing the system to monitor the storage conditions of the medication effectively. Any significant deviations from safe levels triggered alerts on the LCD and with the buzzer, ensuring that users were notified promptly.
    - The IR sensor successfully detected when a pill was dispensed. This feature provided immediate feedback to the user, confirming that the system was operating as intended.
3. **User Interaction**:
    - The LCD display effectively communicated essential information, such as the current time, temperature, humidity, and alerts for medication dispensing. Users found the interface easy to understand and interact with, which is crucial for adherence to medication schedules.
    - The buzzer served as an effective alert mechanism, drawing attention when necessary. Users reported that the sound notifications were helpful for reminding them to take their medication.
4. **Safety Features**:
    - The tilt sensor successfully detected when the device was tilted or moved, which paused the dispensing mechanism to prevent accidental releases. This feature is particularly important for safety, especially in homes with children or pets.
5. **System Reliability**:
    - Throughout the testing phase, the system demonstrated consistent performance without any crashes or errors. It operated reliably over extended periods, suggesting that the hardware and software design is robust.
6. **Limitations**:
    - One limitation noted during testing was the DHT11 sensor's range, which may not cover extreme environmental conditions. Future iterations may benefit from using a more advanced sensor with a wider range.
    - The system currently relies on a set time for dispensing pills. Adding a feature for user input to adjust dispensing times could enhance flexibility.
7. **User Feedback**:
    - Feedback from test users was positive. Many expressed that the system made it easier to manage their medication schedules and reduce the risk of missed doses. Suggestions for improvement included a more user-friendly way to input dosing times and enhancing the visual alerts on the LCD.

Conclusion

The smart pill dispensing system proved to be an effective and reliable solution for medication management. The successful integration of various sensors and components resulted in a user-friendly device that enhances adherence to medication schedules while ensuring safety and ease of use. Future developments could focus on improving sensor capabilities and expanding user interaction features to make the system even more versatile. Overall, the project achieved its objectives, demonstrating the potential of automated systems in healthcare management

CHAPTER 8.

CONCLUSIONS

The smart pill dispensing system successfully meets its objectives of facilitating medication management and enhancing user compliance. Through the integration of various sensors and components, the system operates effectively, providing accurate dispensing notifications, environmental monitoring, and user alerts. Key conclusions drawn from the project include:

1. **Effective Medication Management**: The system ensures that users receive their medication on time, reducing the likelihood of missed doses and improving adherence to prescribed schedules.
2. **Reliable Sensors**: The DHT11 sensor effectively monitored environmental conditions, while the IR and PIR sensors provided accurate feedback on pill dispensing and user presence, respectively. This combination enhances the overall functionality of the device.

3. **User-Friendly Interface**: The LCD display and buzzer notifications provided clear and immediate information to users, making the system easy to operate and understand.

4. **Safety Features**: The inclusion of a tilt sensor contributes to the system's safety, preventing accidental dispensing and ensuring that the device functions as intended in various situations.

5. **Positive User Feedback**: Test users reported high satisfaction with the system, highlighting its effectiveness in managing medication schedules and suggesting areas for improvement, such as user input flexibility.

Overall, the project demonstrates the potential of smart automation in healthcare, paving the way for further advancements in medication management technologies. Future iterations could focus on enhancing sensor capabilities, user interaction, and customization options to further improve the system's utility and user experience.

CHAPTER 9.

FUTURE SCOPE

The smart pill dispensing system has significant potential for further development and enhancement. Here are some areas of future scope:

1. **Advanced Sensors**:
    - Upgrading to more sophisticated sensors, such as the DHT22 or other environmental sensors, could provide better accuracy in temperature and humidity measurements. This would enhance the monitoring capabilities for medication storage conditions.
2. **Mobile Application Integration**:
    - Developing a mobile application could allow users to remotely manage their medication schedules, receive notifications, and monitor system status in real-time. Integration with smartphone alerts could further improve user compliance.
3. **User Input Flexibility**:
    - Implementing a user-friendly interface for setting and modifying dispensing times directly on the LCD or through a mobile app would provide greater customization. This feature would enable users to adjust schedules according to their specific needs.
4. **Data Logging and Analysis**:
    - Incorporating data logging features could track medication intake patterns, environmental conditions, and user interactions over time. This data could be analyzed to provide insights into medication adherence and effectiveness.
5. **Voice Assistant Integration**:
    - Integrating voice recognition technology or compatibility with existing voice assistants (like Amazon Alexa or Google Assistant) could offer users a hands-free way to interact with the system and receive reminders.
6. **Remote Monitoring for Caregivers**:
    - Developing features for caregivers to monitor the medication adherence of patients remotely could enhance healthcare management for the elderly or those with chronic illnesses, providing peace of mind to both patients and their families.
7. **Improved Alert Systems**:

- Enhancing alert mechanisms to include visual signals (such as LED lights) and more customizable sound notifications would make the system more effective for users with varying needs.

8. **Multi-Medication Support**:
    - Expanding the system's capacity to manage multiple medications would cater to users with complex medication regimens, allowing them to manage various prescriptions from a single device.
9. **Integration with Healthcare Systems**:
    - Establishing a connection with healthcare providers or pharmacy systems could allow for automated refill reminders, prescription updates, and better communication between patients and healthcare professionals.

Conclusion

The future scope of the smart pill dispensing system is vast, with numerous opportunities for improvement and expansion. By incorporating advanced technologies and user-centric features, the system can evolve into a comprehensive solution for medication management, ultimately enhancing user adherence and improving health outcomes.

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